

ECRM Report Query Documentation

SQL Schema Reference & Data Dictionary

Database: ecrm_prod | Schema: ecrm, apsis_data | Engine: PostgreSQL (AWS RDS)

Reports Covered

1. Daily Contact Survey Report
2. Live Observation Report (Joint Call)
3. Call Checkback Report
4. Call Center Report

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1. Daily Contact Survey Report

1.1 Report Overview

Field	Details
Report Name	Daily Contact Survey Report
Source Table	ecrm.contacts (c)
Filter / Sample ID	c.id = 12429776 (remove for full data set)
Timezone	UTC stored → Asia/Dhaka (UTC+6) for display
Pivot Method	CASE WHEN + MAX() — one row per contact

1.2 Purpose

Generates a comprehensive per-contact record combining contact metadata, full location hierarchy (Outlet to Cluster to Route to Distributor Point to Territory to Area to Region), consumer profile, brand information, timing, and pivoted survey answers. Each row represents one unique contact.

1.3 Location Hierarchy (Bottom to Top)

Alias	Table	Type	Output Column	Notes
r	ecrm.locations	1	Region	Highest geography level
l_area	ecrm.locations	2	Area	Under Region
l_territory	ecrm.locations	3	Territory	Under Area
l_point	ecrm.locations	5	Distributorspoint	Distributor point
l_route	ecrm.locations	6	Routes	Under Distributor point
l_cluster	ecrm.locations	7	Cluster Name	Under Route
l_outlet	ecrm.locations	8	Outlets Name / Location ID	Lowest — join anchor

1.4 Output Columns — Full Schema Mapping

Output Column	Source Expression	Table	Notes
Contact ID	c.id	ecrm.contacts	PRIMARY KEY — unique contact identifier
Location ID	l_outlet.id	ecrm.locations (type=8)	Outlet location PK
Region	r.name	ecrm.locations (type=1)	Top-level geography
Area	l_area.name	ecrm.locations (type=2)	—
Territory	l_territory.name	ecrm.locations (type=3)	—
Distributorspoint	l_point.name	ecrm.locations (type=5)	—
Routes	l_route.name	ecrm.locations (type=6)	—
Cluster Name	l_cluster.name	ecrm.locations (type=7)	—
Outlets Name	l_outlet.name	ecrm.locations (type=8)	—
User Name	c.contacted_br	ecrm.contacts	BR username who conducted survey
BR Name	ui.full_name	ecrm.user_infos	Full name of BR
Campaign Name	ca.name	ecrm.campaigns	—
Contact Date	c.contact_date	ecrm.contacts	—
Consumer Number	c.contact_no	ecrm.contacts	Phone number (join key to consumers)
Consumer Name	co.name	ecrm.consumers	—
Consumers fathers_name	co.fathers_name	ecrm.consumers	—
AGE	AGE(current_date, co.dob)	ecrm.consumers	Calculated — interval type
Consumer Address	co.address	ecrm.consumers	—
Consumer Occupation	co.profession	ecrm.consumers	—
Primary Brand	pr.name	ecrm.products	c.product → products.id
Secondary Brand	sr.name	ecrm.products	c.secondary_brand → products.id
Contact Start	TO_CHAR(c.start AT TIME ZONE 'Asia/Dhaka', 'HH24:MI:SS')	ecrm.contacts	Timezone-converted start time
Contact End	TO_CHAR(c.end AT TIME ZONE 'Asia/Dhaka', 'HH24:MI:SS')	ecrm.contacts	Timezone-converted end time
Contact Duration	(c.end - c.start)	ecrm.contacts	Interval = end minus start
Contact Picture / signature	c.signature	ecrm.contacts	S3 URL or filename

asking_about_kit	cs.answer (pivot)	ecrm.contact_survey_data_maps	Survey question pivot
asking_about_music_kit_preference	cs.answer (pivot)	ecrm.contact_survey_data_maps	—
asking_about_travel_kit_preference	cs.answer (pivot)	ecrm.contact_survey_data_maps	—
asking_about_art_kit_preference	cs.answer (pivot)	ecrm.contact_survey_data_maps	—
social_media_info	cs.answer (pivot)	ecrm.contact_survey_data_maps	—

1.5 Join Map

Table	Alias	Join Condition / Role
ecrm.contacts	c	Primary source — all contact records
ecrm.locations (type=8)	l_outlet	INNER JOIN — outlet; c.location_id = l_outlet.id
ecrm.locations (type=7)	l_cluster	LEFT JOIN — parent of outlet
ecrm.locations (type=6)	l_route	LEFT JOIN — parent of cluster
ecrm.locations (type=5)	l_point	LEFT JOIN — parent of route
ecrm.locations (type=3)	l_territory	LEFT JOIN — parent of point
ecrm.locations (type=2)	l_area	LEFT JOIN — parent of territory
ecrm.locations (type=1)	r	LEFT JOIN — parent of area (Region)
ecrm.user_infos	ui	LEFT JOIN — c.user_id = ui.user_id
ecrm.campaigns	ca	LEFT JOIN — c.campaign_id = ca.id
ecrm.consumers	co	LEFT JOIN — c.contact_no = co.contact_no
ecrm.products	pr	LEFT JOIN — c.product = pr.id (primary brand)
ecrm.products	sr	LEFT JOIN — c.secondary_brand = sr.id
ecrm.contact_survey_data_maps	cs	INNER JOIN — c.id = cs.contact_id

1.6 Pivot Logic

Survey answers are stored as key-value rows in ecrm.contact_survey_data_maps (one row per question per contact). The query uses CASE WHEN + MAX() to pivot these rows into columns — without aggregation one contact would produce multiple rows. The Report_Download variant uses separate LEFT JOINs per question (cs1, cs2, cs3) as an alternative pivot approach.

1.7 Common Modifications

Modification	How
Remove single-ID filter	DELETE: WHERE c.id = 12429776
Date range filter	ADD: AND c.contact_date BETWEEN '2025-01-01' AND '2025-12-31'
Campaign filter	ADD: AND ca.name = 'Campaign Name'
Add survey question	ADD: MAX(CASE WHEN cs.question = 'your_question' THEN cs.answer END) AS "your_question" — also add to GROUP BY if not aggregated
Order results	ADD at end: ORDER BY c.contact_date DESC
Limit rows	ADD at end: LIMIT 100

1.8 Full Query

The complete production query for the Daily Contact Survey Report. Copy and run directly against ecrm_prod via the read replica.

```

SELECT
  -- CONTACT IDENTIFIER
  c.id                AS "Contact ID",
  l_outlet.id         AS "Location ID",

  -- LOCATION HIERARCHY (Region → Outlet)
  r.name              AS "Region",
  l_area.name         AS "Area",
  l_territory.name    AS "Territory",
  l_point.name        AS "Distributorspoint",
  l_route.name        AS "Routes",
  l_cluster.name      AS "Cluster Name",
  l_outlet.name       AS "Outlets Name",

  -- USER & CAMPAIGN
  c.contacted_br      AS "User Name",
  ui.full_name        AS "BR Name",
  ca.name             AS "Campaign Name",

  -- CONTACT DETAILS
  c.contact_date      AS "Contact Date",
  c.contact_no        AS "Consumer Number",

  -- CONSUMER PROFILE
  co.name             AS "Consumer Name",
  co.fathers_name      AS "Consumers fathers_name",
  AGE(current_date, co.dob) AS "AGE",
  co.address          AS "Consumer Address",
  co.profession       AS "Consumer Occupation",

  -- BRAND
  pr.name             AS "Primary Brand",
  sr.name             AS "Secondary Brand",

  -- TIMING (UTC → Asia/Dhaka)

```

```

    TO_CHAR(c.start AT TIME ZONE 'Asia/Dhaka','HH24:MI:SS') AS "Contact Start",
    TO_CHAR(c.end AT TIME ZONE 'Asia/Dhaka','HH24:MI:SS') AS "Contact End",
    (c.end - c.start) AS "Contact Duration",
    c.signature AS "Contact Picture",

    -- SURVEY ANSWERS (PIVOT)
    MAX(CASE WHEN cs.question='asking_about_kit'
              THEN cs.answer END) AS "asking_about_kit",
    MAX(CASE WHEN cs.question='asking_about_music_kit_preference'
              THEN cs.answer END) AS "asking_about_music_kit_preference",
    MAX(CASE WHEN cs.question='asking_about_travel_kit_preference'
              THEN cs.answer END) AS "asking_about_travel_kit_preference",
    MAX(CASE WHEN cs.question='asking_about_art_kit_preference'
              THEN cs.answer END) AS "asking_about_art_kit_preference",
    MAX(CASE WHEN cs.question='social_media_info'
              THEN cs.answer END) AS "social_media_info"

FROM ecrm.contacts c

    -- OUTLET (type=8) - INNER JOIN required for location chain
    JOIN ecrm.locations l_outlet
      ON c.location_id = l_outlet.id AND l_outlet.type = 8

    -- CLUSTER → ROUTE → POINT → TERRITORY → AREA → REGION
    LEFT JOIN ecrm.locations l_cluster ON l_outlet.parent = l_cluster.id AND l_cluster.type =
7
    LEFT JOIN ecrm.locations l_route ON l_cluster.parent = l_route.id AND l_route.type
= 6
    LEFT JOIN ecrm.locations l_point ON l_route.parent = l_point.id AND l_point.type
= 5
    LEFT JOIN ecrm.locations l_territory ON l_point.parent = l_territory.id AND
l_territory.type = 3
    LEFT JOIN ecrm.locations l_area ON l_territory.parent = l_area.id AND l_area.type
= 2
    LEFT JOIN ecrm.locations r ON l_area.parent = r.id AND r.type
= 1

    LEFT JOIN ecrm.user_infos ui ON c.user_id = ui.user_id
    LEFT JOIN ecrm.campaigns ca ON c.campaign_id = ca.id
    LEFT JOIN ecrm.consumers co ON c.contact_no = co.contact_no
    LEFT JOIN ecrm.products pr ON c.product = pr.id
    LEFT JOIN ecrm.products sr ON c.secondary_brand = sr.id

    -- SURVEY - all questions in one join, pivoted in SELECT
    JOIN ecrm.contact_survey_data_maps cs ON c.id = cs.contact_id

WHERE c.id = 12429776 -- remove for all contacts

GROUP BY
    c.id, l_outlet.id, r.name, l_area.name, l_territory.name,
    l_point.name, l_route.name, l_cluster.name, l_outlet.name,
    c.contacted_br, ui.full_name, ca.name, c.contact_date,
    c.contact_no, co.name, co.fathers_name, co.dob,
    co.address, co.profession, pr.name, sr.name,
    c.start, c.end, c.signature;

```

1.9 Query Clause-by-Clause Explanation

Each significant clause or expression explained in plain language:

SQL Clause / Expression	What It Does
SELECT c.id AS "Contact ID"	The unique primary key of ecrm.contacts. Every row in the result represents exactly one contact event identified by this ID.
l_outlet.id AS "Location ID"	The outlet location's PK (type=8). Captured separately from the outlet name so downstream systems can do ID-based lookups.
r.name, l_area.name ... l_outlet.name	Seven location columns assembled by traversing the parent chain upward from the outlet. Each level is a separate self-join on ecrm.locations filtered by its type number (1=Region, 2=Area, 3=Territory, 5=Point, 6=Route, 7=Cluster, 8=Outlet).
c.contacted_br AS "User Name"	The raw username field stored directly on the contact record — the BR who physically conducted the survey. Does not require a join.
ui.full_name AS "BR Name"	The human-readable full name of the same BR, pulled from ecrm.user_infos via c.user_id to ui.user_id.
AGE(current_date, co.dob)	PostgreSQL built-in that returns an interval (e.g. '28 years 10 mons 7 days'). Calculated fresh at query time — not stored — so it always reflects the consumer's current age.
TO_CHAR(c.start AT TIME ZONE 'Asia/Dhaka', 'HH24:MI:SS')	The contact start timestamp is stored in UTC. AT TIME ZONE 'Asia/Dhaka' shifts it to UTC+6 (Bangladesh Standard Time). TO_CHAR then formats it as HH24:MI:SS for display.
(c.end - c.start) AS "Contact Duration"	Interval arithmetic: PostgreSQL subtracts two timestamps and returns an interval such as '00:06:43'. No function call needed — the minus operator works directly on timestamp columns.
MAX(CASE WHEN cs.question='...' THEN cs.answer END)	Pivot pattern. ecrm.contact_survey_data_maps stores one row per question per contact. Without this, a contact with 5 questions returns 5 rows. CASE WHEN returns the answer only when the question matches (NULL otherwise), and MAX() collapses all rows for the same contact into a single row, picking the one non-NULL value per question column.
FROM ecrm.contacts c	The driving (left-most) table. Every row here is a potential result row. All joins hang off this table.
JOIN ecrm.locations l_outlet ON ... AND l_outlet.type = 8	INNER JOIN — contacts without a matching outlet are excluded entirely. The type=8 filter ensures we only match the Outlet level, not any other location level accidentally sharing the same ID.
LEFT JOIN ecrm.locations l_cluster ON l_outlet.parent = l_cluster.id AND l_cluster.type = 7	Walks one level up the location tree. parent is a foreign key pointing to the next-higher location. LEFT JOIN means if an outlet has no cluster parent, the contact row still appears (cluster columns are NULL).
LEFT JOIN ecrm.consumers co ON c.contact_no = co.contact_no	Consumer profile is joined by phone number, not by ID. This means if the same phone number has multiple consumer records, multiple rows may be returned — worth checking for duplicates on production data.
LEFT JOIN ecrm.products pr ON c.product = pr.id	Two separate joins on ecrm.products with different aliases (pr for primary, sr for secondary brand). Both resolve the integer brand ID stored on the contact to a human-readable product name.
JOIN ecrm.contact_survey_data_maps cs ON c.id = cs.contact_id	INNER JOIN — contacts with no survey answers are excluded. One survey row per question, so this multiplies rows before GROUP BY collapses them again via MAX(CASE WHEN).

WHERE c.id = 12429776	Development/testing filter. Remove this line (or replace with a date range) to run the report for all contacts. Without a filter the query may return millions of rows.
GROUP BY c.id, l_outlet.id, ...	Required because MAX() is an aggregate function. Every SELECT column that is not inside an aggregate must appear here. The GROUP BY key is effectively c.id — all other columns are functionally dependent on it, but PostgreSQL requires them listed explicitly.



2. Live Observation Report (Joint Call)

2.1 Report Overview

Field	Details
Report Name	Live Observation Report (Joint Call)
Source Table	ecrm.joint_calls (jc)
Filter / Sample ID	jc.id = 444487 (remove for full data set)
Supervisor	jc.user_id → ecrm.users (sup)
BR / FF	jc.ff_id → ecrm.users (br)
Pivot Method	CASE WHEN + MAX() — one row per joint call

2.2 Purpose

Records a supervisor's live observation of a BR (Field Force) conducting a consumer contact. The supervisor evaluates whether the BR followed the correct process: correct cluster, correct attire, shop owner permission, device compliance, correct consumer profile, and full campaign messaging execution. Each row = one joint call observation.

2.3 Cross-Schema Join

This report joins across two database schemas: ecrm (primary CRM data) and apsis_data (distributor/company master data). The join key is `l_point.name = dp.name` — a name-based match rather than an ID. Ensure point names are consistent across both schemas to avoid missing rows.

2.4 Output Columns — Full Schema Mapping

Output Column	Source Expression	Table	Notes
Contact ID	jc.id	ecrm.joint_calls	PRIMARY KEY
Region	r.name	ecrm.locations (type=1)	—
Area	l_area.name	ecrm.locations (type=2)	—
Territory	l_territory.name	ecrm.locations (type=3)	—
Distributorspoint	l_point.name	ecrm.locations (type=5)	—
Campaign Name	ca.name	ecrm.campaigns	—
Sup ID	sup.username	ecrm.users	Supervisor username
Sup Name	sup_i.full_name	ecrm.user_infos	Supervisor full name
BR ID	br.username	ecrm.users	BR username — from jc.ff_id
BR Name	br_i.full_name	ecrm.user_infos	BR full name
Contact Date	jc.contact_date	ecrm.joint_calls	—
ra_correct_cluster	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	Survey pivot
ra_correct_attire	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
shop_owner_permission	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—

exists_third_part_app	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
choosing_correct_consumer_profile	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
telling_about_lucky_strike	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
showing_av	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
telling_about_circle	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
telling_about_circle2	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
telling_about_circle3	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
showing_Global_stories	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	Note: mixed-case key
telling_never_ordinary_story	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
asking_to_color	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
showing_kit	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
telling_about_sticker_pack	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—
telling_about_circle_new_journey	jcs.answer (pivot)	ecrm.joint_calls_survey_data_maps	—

2.5 Join Map

Table	Alias	Join Condition / Role
ecrm.joint_calls	jc	Primary source — supervisor joint call records
ecrm.locations (type=8)	l_outlet	LEFT JOIN — jc.location_id = l_outlet.id
ecrm.locations (type=7)	l_cluster	LEFT JOIN — l_outlet.parent = l_cluster.id
ecrm.locations (type=6)	l_route	LEFT JOIN — l_cluster.parent = l_route.id
ecrm.locations (type=5)	l_point	LEFT JOIN — l_route.parent = l_point.id
ecrm.locations (type=3)	l_territory	LEFT JOIN — l_point.parent = l_territory.id
ecrm.locations (type=2)	l_area	LEFT JOIN — l_territory.parent = l_area.id
ecrm.locations (type=1)	r	LEFT JOIN — l_area.parent = r.id (Region)
ecrm.campaigns	ca	LEFT JOIN — jc.campaign_id = ca.id
apsis_data.distributorspoint	dp	LEFT JOIN — l_point.name = dp.name (cross-schema)

apsis_data.company	co	LEFT JOIN — dp.dsid = co.id
ecrm.users	sup	LEFT JOIN — jc.user_id = sup.id (Supervisor)
ecrm.user_infos	sup_i	LEFT JOIN — sup.id = sup_i.user_id
ecrm.users	br	LEFT JOIN — jc.ff_id = br.id (BR/FF)
ecrm.user_infos	br_i	LEFT JOIN — br.id = br_i.user_id
ecrm.joint_calls_survey_data_maps	jcs	LEFT JOIN — jc.id = jcs.joint_call_id

2.6 Key Differences from Daily Report

Aspect	Daily Report	Live Observation
Primary table	ecrm.contacts	ecrm.joint_calls
BR field	c.user_id	jc.ff_id
Supervisor	Not present	jc.user_id → sup
Survey table	contact_survey_data_maps	joint_calls_survey_data_maps
Consumer data	Full profile (name, age, brand...)	Not captured
Cross-schema	No	Yes — apsis_data

2.7 Full Query

The complete production query for the Live Observation (Joint Call) Report.

```

SELECT
    jc.id                                AS "Contact ID",

    -- LOCATION HIERARCHY
    r.name                              AS "Region",
    l_area.name                          AS "Area",
    l_territory.name                     AS "Territory",
    l_point.name                         AS "Distributorspoint",

    -- CAMPAIGN
    ca.name                              AS "Campaign Name",

    -- SUPERVISOR
    sup.username                         AS "Sup ID",
    sup_i.full_name                      AS "Sup Name",

    -- BR (FIELD FORCE)
    br.username                          AS "BR ID",
    br_i.full_name                       AS "BR Name",

    jc.contact_date                      AS "Contact Date",

    -- SURVEY ANSWERS (PIVOT)
    MAX(CASE WHEN jcs.question='ra_correct_cluster' THEN jcs.answer END) AS
    "ra_correct_cluster",
    MAX(CASE WHEN jcs.question='ra_correct_attire' THEN jcs.answer END) AS
    "ra_correct_attire",
    MAX(CASE WHEN jcs.question='shop_owner_permission' THEN jcs.answer END) AS
    "shop_owner_permission",
    MAX(CASE WHEN jcs.question='exists_third_part_app' THEN jcs.answer END) AS
    "exists_third_part_app",

```

```

    MAX(CASE WHEN jcs.question='choosing_correct_consumer_profile' THEN jcs.answer END) AS
"choosing_correct_consumer_profile",
    MAX(CASE WHEN jcs.question='telling_about_lucky_strike' THEN jcs.answer END) AS
"telling_about_lucky_strike",
    MAX(CASE WHEN jcs.question='showing_av' THEN jcs.answer END) AS
"showing_av",
    MAX(CASE WHEN jcs.question='telling_about_circle' THEN jcs.answer END) AS
"telling_about_circle",
    MAX(CASE WHEN jcs.question='telling_about_circle2' THEN jcs.answer END) AS
"telling_about_circle2",
    MAX(CASE WHEN jcs.question='telling_about_circle3' THEN jcs.answer END) AS
"telling_about_circle3",
    MAX(CASE WHEN jcs.question='showing_Global_stories' THEN jcs.answer END) AS
"showing_Global_stories",
    MAX(CASE WHEN jcs.question='telling_never_ordinary_story' THEN jcs.answer END) AS
"telling_never_ordinary_story",
    MAX(CASE WHEN jcs.question='asking_to_color' THEN jcs.answer END) AS
"asking_to_color",
    MAX(CASE WHEN jcs.question='showing_kit' THEN jcs.answer END) AS
"showing_kit",
    MAX(CASE WHEN jcs.question='telling_about_sticker_pack' THEN jcs.answer END) AS
"telling_about_sticker_pack",
    MAX(CASE WHEN jcs.question='telling_about_circle_new_journey' THEN jcs.answer END) AS
"telling_about_circle_new_journey"

FROM ecrm.joint_calls jc

-- LOCATION CHAIN (Outlet → Region)
LEFT JOIN ecrm.locations l_outlet ON jc.location_id = l_outlet.id AND
l_outlet.type = 8
LEFT JOIN ecrm.locations l_cluster ON l_outlet.parent = l_cluster.id AND
l_cluster.type = 7
LEFT JOIN ecrm.locations l_route ON l_cluster.parent = l_route.id AND
l_route.type = 6
LEFT JOIN ecrm.locations l_point ON l_route.parent = l_point.id AND
l_point.type = 5
LEFT JOIN ecrm.locations l_territory ON l_point.parent = l_territory.id AND
l_territory.type = 3
LEFT JOIN ecrm.locations l_area ON l_territory.parent = l_area.id AND l_area.type
= 2
LEFT JOIN ecrm.locations r ON l_area.parent = r.id AND r.type
= 1

LEFT JOIN ecrm.campaigns ca ON jc.campaign_id = ca.id

-- CROSS-SCHEMA: apsis_data (name-based match)
LEFT JOIN apsis_data.distributorspoint dp ON l_point.name = dp.name
LEFT JOIN apsis_data.company co ON dp.dsid = co.id

-- SUPERVISOR
LEFT JOIN ecrm.users sup ON jc.user_id = sup.id
LEFT JOIN ecrm.user_infos sup_i ON sup.id = sup_i.user_id

-- BR / FIELD FORCE (jc.ff_id - different from daily report's user_id)
LEFT JOIN ecrm.users br ON jc.ff_id = br.id
LEFT JOIN ecrm.user_infos br_i ON br.id = br_i.user_id

LEFT JOIN ecrm.joint_calls_survey_data_maps jcs ON jc.id = jcs.joint_call_id

WHERE jc.id = 444487 -- remove for all joint calls

GROUP BY
jc.id, r.name, l_area.name, l_territory.name, l_point.name,
ca.name, sup.username, sup_i.full_name,
br.username, br_i.full_name, jc.contact_date;

```

2.8 Query Clause-by-Clause Explanation

SQL Clause / Expression	What It Does
<code>FROM ecrm.joint_calls jc</code>	The primary table for this report. Each row is one supervisor joint call — a live field visit where the supervisor accompanies and evaluates a BR. Not the same as a BR's own contact record.
<code>LEFT JOIN ecrm.locations l_outlet ON jc.location_id = l_outlet.id AND l_outlet.type = 8</code>	Unlike the Daily Report which uses INNER JOIN for the outlet, this report uses LEFT JOIN throughout the entire location chain. A joint call record missing a location will still appear — location columns will be NULL.
<code>l_outlet.parent = l_cluster.id AND l_cluster.type = 7</code>	The self-referential join pattern: every location record has a parent column pointing to the next level up. The type filter (7, 6, 5, 3, 2, 1) ensures each alias only matches its intended level — without it, a single location ID could match multiple levels.
<code>LEFT JOIN apsis_data.distributorspoint dp ON l_point.name = dp.name</code>	Cross-schema join to the apsis_data database. Crucially, this joins on name (text), not on an ID. If a distributor point name is spelled differently in the two schemas (spaces, capitalisation, trailing characters), the join silently returns NULL. Always verify name consistency when distributor data is missing.
<code>LEFT JOIN apsis_data.company co ON dp.dsid = co.id</code>	Second cross-schema join — follows from the distributorspoint join to resolve the parent company. If dp is NULL (name mismatch above), co will also be NULL automatically.
<code>LEFT JOIN ecrm.users sup ON jc.user_id = sup.id</code>	jc.user_id identifies the SUPERVISOR who conducted the observation — different from the Daily Report where user_id is the BR. This semantic difference is critical: swapping the fields would attribute supervisor actions to the BR.
<code>LEFT JOIN ecrm.users br ON jc.ff_id = br.id</code>	jc.ff_id (Field Force ID) identifies the BR being observed. The field name ff_id is the only way to distinguish supervisor from BR in this table — there is no 'role' column.
<code>jcs.question = 'showing_Global_stories'</code>	WARNING: This question key uses mixed case — capital G in 'Global'. The equivalent key in the Call Checkback report uses all-lowercase 'showing_global_series'. If you filter or search by question key across reports, match the exact case for each report.
<code>LEFT JOIN ecrm.joint_calls_survey_data_maps jcs ON jc.id = jcs.joint_call_id</code>	Survey data for joint calls is stored in a separate table from the Daily Report's contact_survey_data_maps. The join column is joint_call_id (not contact_id). Using the wrong survey table would silently return no survey answers.
<code>WHERE jc.id = 444487</code>	Development filter for a single joint call. Remove for production runs. Consider adding a date filter: AND jc.contact_date = '2025-07-14'.
<code>GROUP BY jc.id, r.name, ..., jc.contact_date</code>	All non-aggregated SELECT columns must appear here. The minimal unique key is jc.id, but PostgreSQL requires every column listed. The contact_date is included because it may vary per joint_call_id in edge cases.

3. Call Checkback Report

3.1 Report Overview

Field	Details
Report Name	Call Checkback Report (Supervisor Contact Verification)
Source Table	ecrm.supervisor_contacts (sc)
Filter / Sample ID	sc.id = 495011 (remove for full data set)
Supervisor	sc.user_id → ecrm.users (sup)
BR	sc.br_id → ecrm.users (br)
Contact Match	sc.contact + sc.br_id + sc.campaign_id + sc.contact_date = ecrm.contacts
Pivot Method	CASE WHEN + MAX() — one row per supervisor contact

3.2 Purpose

A supervisor calls back consumers after a BR contact to verify whether the consumer actually received the call, whether they were willing to talk, and whether the BR correctly communicated all campaign messages. This is a post-contact quality audit — the supervisor validates the BR's earlier work.

3.3 Four-Key Contact Match

Unlike the other reports, this report links supervisor contacts back to original BR contacts via a composite four-column join on ecrm.contacts — matching contact phone number, BR ID, campaign ID, and contact date together. This is critical: a mismatch on any one of the four keys means no location data will appear for that row.

3.4 Output Columns — Full Schema Mapping

Output Column	Source Expression	Table	Notes
Contact ID	sc.id	ecrm.supervisor_contacts	PRIMARY KEY of supervisor contact
Region	re.name	ecrm.locations (type=1)	—
Area	ar.name	ecrm.locations (type=2)	—
Territory	ter.name	ecrm.locations (type=3)	—
Distributorspoint	po.name	ecrm.locations (type=5)	—
Campaign Name	cm.name	ecrm.campaigns	—
Sup ID	sup.username	ecrm.users	Supervisor username
Sup Name	sup_i.full_name	ecrm.user_infos	—

BR ID	br.username	ecrm.users	BR — from sc.br_id
BR Name	br_i.full_name	ecrm.user_infos	—
Contact Date	sc.contact_date	ecrm.supervisor_contacts	—
receive_call	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	Survey pivot
continue_conversation	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
correct_consumer_name	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
correct_consumer_age	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
correct_primary_brand	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
contacted_by_ra	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
telling_about_lucky_strike	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
showing_av	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
telling_about_circle	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
telling_about_circle2	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
telling_about_circle3	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
showing_global_series	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	Note: lowercase 'g' (vs Live Obs. 'G')
telling_never_ordinary_story	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
asking_to_color	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
showing_kit	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
telling_about_sticker_pack	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—
telling_about_circle_new_journey	scs.answer (pivot)	ecrm.supervisor_contact_survey_data_ma ps	—

3.5 Join Map

Table	Alias	Join Condition / Role
ecrm.supervisor_contacts	sc	Primary source — supervisor verification records
ecrm.contacts	c	LEFT JOIN — 4-key match: contact_no + br_id + campaign_id + contact_date
ecrm.locations (type=8)	ou	LEFT JOIN — c.location_id = ou.id
ecrm.locations (type=7)	cl	LEFT JOIN — ou.parent = cl.id
ecrm.locations (type=6)	ro	LEFT JOIN — cl.parent = ro.id
ecrm.locations (type=5)	po	LEFT JOIN — ro.parent = po.id
ecrm.locations (type=3)	ter	LEFT JOIN — po.parent = ter.id
ecrm.locations (type=2)	ar	LEFT JOIN — ter.parent = ar.id
ecrm.locations (type=1)	re	LEFT JOIN — ar.parent = re.id (Region)
apsis_data.distributorspoint	dp	LEFT JOIN — po.source_id = dp.id (ID-based, not name)
apsis_data.company	co	LEFT JOIN — dp.dsid = co.id
ecrm.users	sup	LEFT JOIN — sc.user_id = sup.id (Supervisor)
ecrm.user_infos	sup_i	LEFT JOIN — sup.id = sup_i.user_id
ecrm.users	br	LEFT JOIN — sc.br_id = br.id (BR)
ecrm.user_infos	br_i	LEFT JOIN — br.id = br_i.user_id
ecrm.campaigns	cm	LEFT JOIN — sc.campaign_id = cm.id
ecrm.supervisor_contact_survey_data_maps	scs	LEFT JOIN — sc.id = scs.sup_contact_id

3.6 apsis_data Join Difference

In this report, the apsis_data join uses po.source_id = dp.id (an ID-based match). In the Live Observation report, the same join uses l_point.name = dp.name (a name-based match). Be aware of this difference when troubleshooting missing distributor data.

3.7 Full Query

The complete production query for the Call Checkback Report.

```

SELECT
    sc.id                                AS "Contact ID",

    -- LOCATION HIERARCHY
    re.name                             AS "Region",
    ar.name                             AS "Area",
    ter.name                            AS "Territory",
    po.name                             AS "Distributorspoint",

    -- CAMPAIGN
    cm.name                             AS "Campaign Name",

    -- SUPERVISOR
    sup.username                        AS "Sup ID",
    sup_i.full_name                     AS "Sup Name",

```



```

-- BR
br.username          AS "BR ID",
br_i.full_name       AS "BR Name",

sc.contact_date      AS "Contact Date",

-- SURVEY ANSWERS (PIVOT)
MAX(CASE WHEN scs.question='receive_call' THEN scs.answer END) AS
"receive_call",
MAX(CASE WHEN scs.question='continue_conversation' THEN scs.answer END) AS
"continue_conversation",
MAX(CASE WHEN scs.question='correct_consumer_name' THEN scs.answer END) AS
"correct_consumer_name",
MAX(CASE WHEN scs.question='correct_consumer_age' THEN scs.answer END) AS
"correct_consumer_age",
MAX(CASE WHEN scs.question='correct_primary_brand' THEN scs.answer END) AS
"correct_primary_brand",
MAX(CASE WHEN scs.question='contacted_by_ra' THEN scs.answer END) AS
"contacted_by_ra",
MAX(CASE WHEN scs.question='telling_about_lucky_strike' THEN scs.answer END) AS
"telling_about_lucky_strike",
MAX(CASE WHEN scs.question='showing_av' THEN scs.answer END) AS
"showing_av",
MAX(CASE WHEN scs.question='telling_about_circle' THEN scs.answer END) AS
"telling_about_circle",
MAX(CASE WHEN scs.question='telling_about_circle2' THEN scs.answer END) AS
"telling_about_circle2",
MAX(CASE WHEN scs.question='telling_about_circle3' THEN scs.answer END) AS
"telling_about_circle3",
MAX(CASE WHEN scs.question='showing_global_series' THEN scs.answer END) AS
"showing_global_series",
MAX(CASE WHEN scs.question='telling_never_ordinary_story' THEN scs.answer END) AS
"telling_never_ordinary_story",
MAX(CASE WHEN scs.question='asking_to_color' THEN scs.answer END) AS
"asking_to_color",
MAX(CASE WHEN scs.question='showing_kit' THEN scs.answer END) AS
"showing_kit",
MAX(CASE WHEN scs.question='telling_about_sticker_pack' THEN scs.answer END) AS
"telling_about_sticker_pack",
MAX(CASE WHEN scs.question='telling_about_circle_new_journey' THEN scs.answer END) AS
"telling_about_circle_new_journey"

FROM ecrm.supervisor_contacts sc

-- LINK BACK TO ORIGINAL BR CONTACT (4-key composite match)
LEFT JOIN ecrm.contacts c
    ON sc.contact = c.contact_no
    AND sc.br_id = c.user_id
    AND sc.campaign_id = c.campaign_id
    AND sc.contact_date = c.contact_date

-- LOCATION VIA CONTACT (Outlet → Region)
LEFT JOIN ecrm.locations ou ON c.location_id = ou.id AND ou.type = 8
LEFT JOIN ecrm.locations cl ON ou.parent = cl.id AND cl.type = 7
LEFT JOIN ecrm.locations ro ON cl.parent = ro.id AND ro.type = 6
LEFT JOIN ecrm.locations po ON ro.parent = po.id AND po.type = 5
LEFT JOIN ecrm.locations ter ON po.parent = ter.id AND ter.type = 3
LEFT JOIN ecrm.locations ar ON ter.parent = ar.id AND ar.type = 2
LEFT JOIN ecrm.locations re ON ar.parent = re.id AND re.type = 1

-- CROSS-SCHEMA: apsis_data (ID-based match — different from Live Obs.)
LEFT JOIN apsis_data.distributorspoint dp ON po.source_id = dp.id
LEFT JOIN apsis_data.company co ON dp.dsid = co.id

-- SUPERVISOR
LEFT JOIN ecrm.users sup ON sc.user_id = sup.id
LEFT JOIN ecrm.user_infos sup_i ON sup.id = sup_i.user_id

-- BR

```

```

LEFT JOIN ecrm.users br ON sc.br_id = br.id
LEFT JOIN ecrm.user_infos br_i ON br.id = br_i.user_id

LEFT JOIN ecrm.campaigns cm ON sc.campaign_id = cm.id

LEFT JOIN ecrm.supervisor_contact_survey_data_maps scs ON sc.id = scs.sup_contact_id

WHERE sc.id = 495011 -- remove for all supervisor contacts

GROUP BY
  sc.id, re.name, ar.name, ter.name, po.name, cm.name,
  sup.username, sup_i.full_name,
  br.username, br_i.full_name,
  sc.contact_date;

```

3.8 Query Clause-by-Clause Explanation

SQL Clause / Expression	What It Does
FROM ecrm.supervisor_contacts sc	The primary table. Each row is one supervisor phone call made to a consumer to verify a BR's earlier contact. This is a post-contact audit table — it does not replace the original ecrm.contacts record.
LEFT JOIN ecrm.contacts c ON sc.contact = c.contact_no AND sc.br_id = c.user_id AND sc.campaign_id = c.campaign_id AND sc.contact_date = c.contact_date	The four-key composite join. sc.contact matches the consumer phone number, sc.br_id matches the BR who made the original contact, sc.campaign_id matches the campaign, and sc.contact_date matches the date. All four must match simultaneously. If any one key is mismatched (e.g. date format difference, different BR), the join returns no contact record and all location columns will be NULL.
LEFT JOIN ecrm.locations ou ON c.location_id = ou.id AND ou.type = 8	Location data comes from ecrm.contacts (c), not from supervisor_contacts (sc) directly. This means if the composite join above failed (c is NULL), the entire location chain will also be NULL — the location is indirectly dependent on the contact link working correctly.
LEFT JOIN apsis_data.distributorspoint dp ON po.source_id = dp.id	KEY DIFFERENCE from Live Observation: this join uses po.source_id = dp.id (integer ID match). The Live Observation report uses l_point.name = dp.name (text name match). The ID-based join here is more reliable and less prone to mismatch from spelling differences.
LEFT JOIN ecrm.users sup ON sc.user_id = sup.id	sc.user_id is the SUPERVISOR — the person making the checkback call. Same pattern as the Live Observation report.
LEFT JOIN ecrm.users br ON sc.br_id = br.id	sc.br_id is the BR whose original contact is being verified. This is also one of the four keys used in the composite contact join above.
LEFT JOIN ecrm.supervisor_contact_survey_data_maps scs ON sc.id = scs.sup_contact_id	Third distinct survey table across the three reports. The join column is sup_contact_id (not contact_id or joint_call_id). Using the wrong survey table would silently return zero survey answers.
scs.question = 'showing_global_series'	All-lowercase in this report. The equivalent key in the Live Observation report is 'showing_Global_stories' (capital G, different suffix). These are the same conceptual question but stored under different keys in different survey tables.
MAX(CASE WHEN scs.question='receive_call' THEN scs.answer END)	The checkback-specific questions (receive_call, continue_conversation, correct_consumer_name, correct_consumer_age, correct_primary_brand, contacted_by_ra) only exist in this report. They verify the factual accuracy of the original contact record.

WHERE sc.id = 495011	Single-record development filter. For production, replace with a date filter such as: WHERE sc.contact_date BETWEEN '2025-07-01' AND '2025-07-31'.
GROUP BY sc.id, re.name, ..., sc.contact_date	Groups the survey rows back into one row per supervisor contact. The minimal key is sc.id but all non-aggregate SELECT columns must be listed. contact_date is on sc (not c) since location may be NULL.



4. Call Center Report

4.1 Purpose

Tracks outbound call center activity — agents contacting consumers to verify invitation receipt, willingness to attend, and event confirmation. Operates on a separate call-center workflow from the field BR contacts. Survey questions are in English (unlike field reports which use Bengali).

4.2 Output Columns

Output Column	Source Expression	Table	Notes
Contact id	cc.id	ecrm (call center table)	PRIMARY KEY
Region	location hierarchy	ecrm.locations	Type 1
Area	location hierarchy	ecrm.locations	Type 2
Territory	location hierarchy	ecrm.locations	Type 3
Distributors Point	location hierarchy	ecrm.locations	Type 5
Route	location hierarchy	ecrm.locations	Type 6
Cluster	location hierarchy	ecrm.locations	Type 7
User ID	user.username	ecrm.users	Call center agent
User Name	user_info.full_name	ecrm.user_infos	—
Agency	agency name	ecrm (agency table)	—
Campaign Name	ca.name	ecrm.campaigns	—
Consumer Number	contact_no	ecrm (call center)	Consumer phone
Contacted Date	contact_date	ecrm (call center)	Date of current call
Last Contacted Date	last_contact_date	ecrm (call center)	Previous contact date
received_phone_call	survey answer	call center survey map	Did consumer answer?
call_not_received_reason	survey answer	call center survey map	Reason if no answer
agreed_to_talk	survey answer	call center survey map	Consumer willingness
not_agreed_to_talk_reason	survey answer	call center survey map	—
invitation_received	survey answer	call center survey map	Event invite confirmed
Contact Start	TO_CHAR(start, 'HH24:MI:SS')	ecrm (call center)	—
Contact End	TO_CHAR(end, 'HH24:MI:SS')	ecrm (call center)	—
Contact Duration	(end - start)	ecrm (call center)	Interval
Contact Status	derived / status field	ecrm (call center)	e.g. Successful

4.3 Survey Question Aliases

Alias / Column Key	Full Question Text
--------------------	--------------------

received_phone_call	Did the consumer receive the call?
call_not_received_reason	Why the consumer didn't receive the call?
agreed_to_talk	Did the consumer want to talk?
not_agreed_to_talk_reason	Why the consumer didn't want to talk?
invitation_received	Invitation received?



Appendix A — Database Connection Details

Parameter	Value
Name	ecrm_replica
Host	ecrm-prod-read-replica.cg4ujvseetk.ap-southeast-1.rds.amazonaws.com
Port	5432
Database	ecrm_prod
User	ahsam
Engine	PostgreSQL (AWS RDS — Read Replica)
Region	ap-southeast-1 (Singapore)

This is a read replica — safe for SELECT queries. Do not run INSERT, UPDATE, or DELETE against this connection.

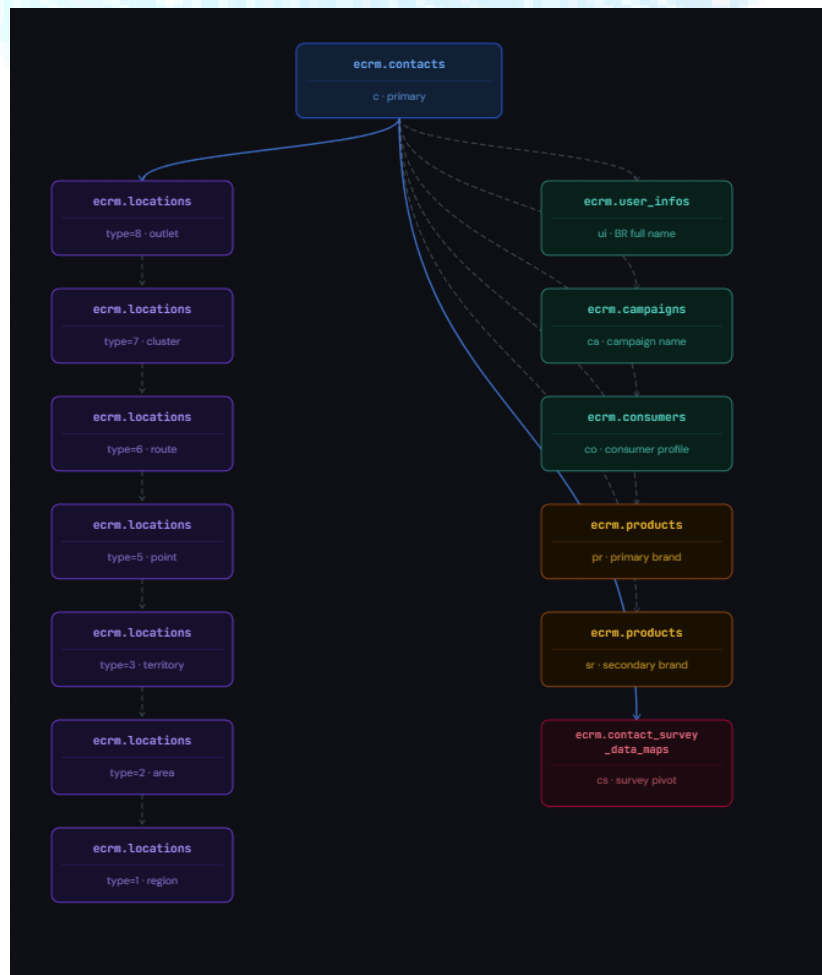
Appendix B — Survey Question Reference (All Reports)

All survey answers are stored in Bengali (transliterated below). Complete alias-to-question mapping across all reports.

Column Alias	Report	Bengali Question (Transliterated)	English Meaning
receive_call	Call Checkback	Consumer ki call receive koreyche?	Did consumer receive call?
continue_conversation	Call Checkback	Consumer ki kotha bolte cheyeche?	Did consumer want to talk?
not_want_to_talk	Call Checkback	Consumer keno kotha bolte icchuk non?	Why not willing to talk?
correct_consumer_name	Call Checkback	Consumer er naam sothik?	Consumer name correct?
correct_consumer_age	Call Checkback	Consumer er boyosh sothik?	Consumer age correct?
correct_primary_brand	Call Checkback	Consumer er bortoman dhumopan brand sothik?	Current smoking brand correct?
contacted_by_ra	Call Checkback	Somproti ki kono RA consumer er shathe jogajog koreche?	Did RA recently contact?
ra_correct_cluster	Live Observation	RA ki sothik cluster e ache?	Is RA in correct cluster?
ra_correct_attire	Live Observation	RA ki sothik attire e ache?	Is RA in correct attire?
shop_owner_permission	Live Observation	RA ki dokandar er kach theke anumati niyo kaj shuru koreche?	Did RA get shop owner permission?
exists_third_part_app	Live Observation	RA er device e ki kono third party apps install kora ache?	Third-party app on device?
choosing_correct_consumer_profile	Live Observation	RA ki sothik profile er consumer nirbacho n koreche?	Correct consumer profile selected?
telling_about_lucky_strike	Both	RA ki consumer ke Lucky Strike somporke boleche?	RA told about Lucky Strike?
showing_av	Both	RA ki consumer ke AV dekhayeche?	RA showed AV?
telling_about_circle	Both	Circle platform somporke RA boleche?	Circle intro messaging
telling_about_circle2	Both	Circle experience detail bola hoyeche?	Circle experience detail
telling_about_circle3	Both	Circle intro tab er madhome dekhano hoyeche?	Tab-based intro comm

showing_global_series / showing_Global_stories	Both (case differs)	Global stories tab er madhome dekhano hoyeche?	Note case difference between reports
telling_never_ordinary_story	Both	Never Ordinary story bola hoyeche?	—
asking_to_color	Both	Consumer ke color korte bola hoyeche?	—
showing_kit	Both	Never Ordinary kit dekhano hoyeche?	—
telling_about_sticker_pack	Both	Sticker pack somporke bolche ebong deoa hoyeche?	—
telling_about_circle_new_journey	Both	New journey messaging bola hoyeche?	—

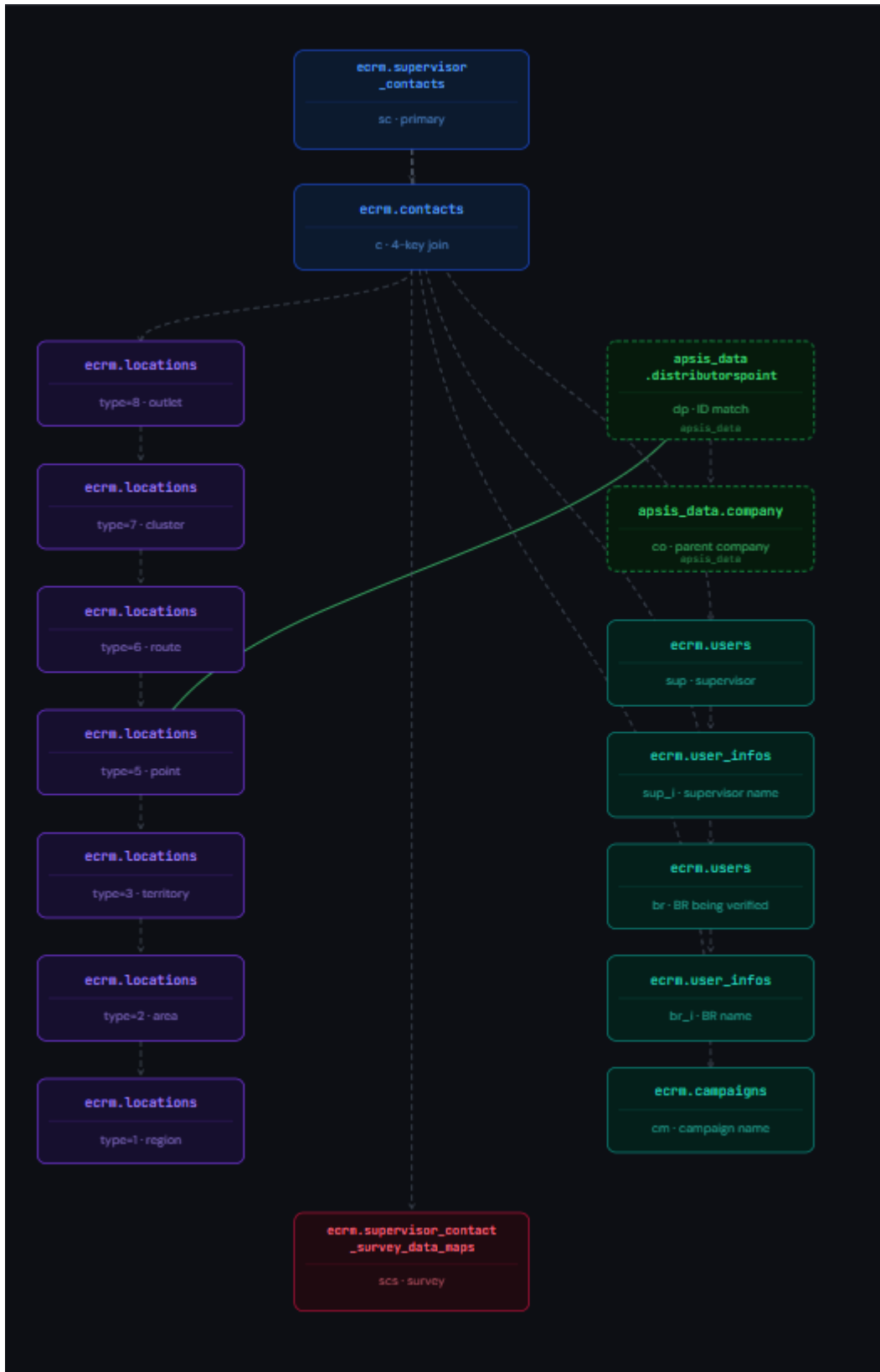
Visual Representation of the Connection ([Daily Contact](#))



Visual Representation of the Connection [\(Live Observation\)](#)



Visual Representation of the Connection ([Call Checkback](#))



Visual Representation of the Connection ([Call Checkback](#))



All Report Visual Together:



